Problem Set 4: CM1 (2026)

4.2 ⋆⋆ Evaluate the work done

$$W = \int_O^P \mathbf{F} \cdot d\mathbf{r} = \int_O^P (F_x\, dx + F_y\, dy) \tag{4.100}$$

by the two-dimensional force $\mathbf{F} = (x^2, 2xy)$ along the three paths joining the origin to the point $P = (1, 1)$ as shown in Figure 4.24(a) and defined as follows: **(a)** This path goes along the x axis to $Q = (1, 0)$ and then straight up to P. (Divide the integral into two pieces, $\int_O^P = \int_O^Q + \int_Q^P$.) **(b)** On this path $y = x^2$, and you can replace the term dy in (4.100) by $dy = 2x\, dx$ and convert the whole integral into an integral over x. **(c)** This path is given parametrically as $x = t^3$, $y = t^2$. In this case rewrite x, y, dx, and dy in (4.100) in terms of t and dt, and convert the integral into an integral over t.

4.9 ⋆⋆ **(a)** The force exerted by a one-dimensional spring, fixed at one end, is $F = -kx$, where x is the displacement of the other end from its equilibrium position. Assuming that this force is conservative (which it is) show that the corresponding potential energy is $U = \frac{1}{2}kx^2$, if we choose U to be zero at the equilibrium position. **(b)** Suppose that this spring is hung vertically from the ceiling with a mass m suspended from the other end and constrained to move in the vertical direction only. Find the extension x_0 of the new equilibrium position with the suspended mass. Show that the total potential energy (spring plus gravity) has the same form $\frac{1}{2}ky^2$ if we use the coordinate y equal to the displacement measured from the new equilibrium position at $x = x_0$ (and redefine our reference point so that $U = 0$ at $y = 0$).

4.14 ⋆ Prove that if $f(\mathbf{r})$ and $g(\mathbf{r})$ are any two scalar functions of $\mathbf{r}$, then

$$\nabla(fg) = f\nabla g + g\nabla f$$

4.18 ⋆⋆ Use the property (4.35) of the gradient to prove the following important results: **(a)** The vector ∇f at any point $\mathbf{r}$ is perpendicular to the surface of constant f through $\mathbf{r}$. (Choose a small displacement $d\mathbf{r}$ that lies in a surface of constant f. What is df for such a displacement?) **(b)** The direction of ∇f at any point $\mathbf{r}$ is the direction in which f increases fastest as we move away from $\mathbf{r}$. (Choose a small displacement $d\mathbf{r} = \epsilon\mathbf{u}$, where $\mathbf{u}$ is a unit vector and ϵ is fixed and small. Find the direction of $\mathbf{u}$ for which the corresponding df is maximum, bearing in mind that $\mathbf{a} \cdot \mathbf{b} = ab\cos\theta$.)

4.23 ⋆⋆ Which of the following forces is conservative? **(a)** $\mathbf{F} = k(x, 2y, 3z)$ where k is a constant. **(b)** $\mathbf{F} = k(y, x, 0)$. **(c)** $\mathbf{F} = k(-y, x, 0)$. For those which are conservative, find the corresponding potential energy U, and verify by direct differentiation that $\mathbf{F} = -\nabla U$.

4.24 ⋆⋆⋆ An infinitely long, uniform rod of mass μ per unit length is situated on the z axis. **(a)** Calculate the gravitational force $\mathbf{F}$ on a point mass m at a distance ρ from the z axis. (The gravitational force between two point masses is given in Problem 4.21.) **(b)** Rewrite $\mathbf{F}$ in terms of the rectangular coordinates (x, y, z) of the point and verify that $\nabla \times \mathbf{F} = 0$. **(c)** Show that $\nabla \times \mathbf{F} = 0$ using the expression for $\nabla \times \mathbf{F}$ in cylindrical polar coordinates given inside the back cover. **(d)** Find the corresponding potential energy U.